

The Foundational Crisis

From Paradox to Program

Jan Reimann

Recap and Roadmap

Last time: Dedekind & Cantor's existential, set-theoretic style of mathematics — and Russell's paradox, which broke Frege's comprehension principle.

Today

- paradoxes and consistency, predicativity, the axiom of choice
- Brouwer's intuitionism vs. Hilbert's program

The question throughout: *what is a legitimate mathematical object, definition, or proof?*

The Comprehension Principle, Again

Recall the principle that broke under Russell's paradox:

for any well-defined property p , there exists $\{x : p(x)\}$

- This was the *basis for the logicist understanding of set theory* — “naive” set theory
- Not just Russell's paradox: **two families** of paradoxes emerged around 1900
 - **set-theoretic** paradoxes (Burali-Forti, Cantor, Russell/Zermelo)
 - **semantic** paradoxes (Richard, König, Grelling) — about truth and definability

The Set-Theoretic Paradoxes

- **Burali-Forti (1897; Cantor 1896):** the set of all ordinals would itself be well-ordered, hence have an ordinal $>$ every ordinal in it — contradiction
- **Cantor (1899):** the set of all cardinals leads to the same kind of contradiction
- **Zermelo–Russell (1901–02):** let $p(x)$ be “ $x \notin x$ ” and consider $R = \{x : x \notin x\}$

Comprehension is contradictory in the *simplest possible way* — no need for a theory of the transfinite to state it.

Hilbert and Existence

Hilbert abandoned logicism. He proposed working with axiom systems that comprise non-logical axioms.

This, however, requires a new kind of justification why the objects we are trying to axiomatize exist.

Hilbert’s proposal:

- A set of objects “exists” only if we can prove its defining axiom system **consistent**
- *concept* $\rightarrow$ *set* becomes *consistency proof* $\rightarrow$ *legitimate existence*
- Objects have a kind of **ideal existence** “in the realm of thought,” not independent metaphysical existence

Predicativity — Poincaré’s Diagnosis

When Frege’s and Russell’s paradoxes became widely known (1903), **Poincaré** offered his own culprit: **impredicative definitions**.

A definition is *impredicative* when it refers to a totality that already contains the very element being defined.

- Example: **least upper bound** – defining the supremum involves quantifying over “all upper bounds,” a set of real numbers, in order to pin down a particular real number. That real number is thus specified by reference to a totality it belongs to.
- Poincaré: this kind of circularity is exactly what licenses the paradoxes

Richard's Paradox (a Semantic Paradox)

- Only countably many reals are *definable* by a finite expression — list them $a_1, a_2, a_3, \dots$ (lexicographic order)
- Define r : the n th decimal digit of r differs from the n th digit of a_n
- r is not on the list — yet r was just *defined*, in finitely many words!

Poincaré's diagnosis: r 's definition refers to the totality of *all* definable numbers, of which r is itself meant to be a member. Impredicative.

Two Responses to Impredicativity

Zermelo — unconvinced

- Impredicative definitions are everywhere in classical analysis (e.g. least upper bound)
- Harmless: the object isn't *created* by the definition, only *singled out* from what already exists

Russell — convinced

- Adopts Poincaré's "vicious circle principle"
- Builds a **theory of types**: objects stratified into levels; no set of "mixed" type
- Result: *Principia Mathematica* (1910–13)

Where Predicativity Led

- Russell's *ramified* type theory needed extra ad hoc machinery (the axiom of **reducibility**) to recover ordinary mathematics
- **Weyl**, in *Das Kontinuum* (1918), tried a leaner predicative foundation: accept classical logic and $\mathbb{N}$, but build the reals predicatively
 - cost: $\mathbb{R}$ is no longer complete — the Bolzano–Weierstrass theorem fails
- Predicative mathematics survives today as a serious *middle ground* between classical and constructive mathematics

Controversy on the Axiom of Choice

Zermelo (1904): $AC \implies$ every set — in particular $\mathbb{R}$ — can be **well-ordered**.

- AC: given any family of disjoint nonempty sets, a *choice set* exists containing one element from each
- The proof gives **no explicit construction** of the well-ordering

- Critics: AC merely *asserts* the choice set exists, without a defining property for it

⇒ a deep debate followed, cutting across the logicism/predicativism divide.

From Controversy to Axiom System

The AC debate spurred Zermelo to work out his **axiom system** for set theory (1908) — a careful, mathematical (not logical) analysis of Cantor’s and Dedekind’s set theory.

- **Fraenkel & von Neumann** (1920s) add: **Replacement** and **Regularity**
- **Weyl & Skolem**: formalize within **first-order logic** — quantify only over *sets*, not over *properties*

$$\leadsto \text{ZF(C)}$$

What ZFC Codifies

ZFC settles the early-1900s controversies by simply **building in** the modern, anti-Kroneckerian traits:

- impredicative definitions — **allowed**
- arbitrary functions and subsets — **allowed**
- purely existential proofs — **allowed**

The result: a working “**Platonistic**” **view** of mathematics — the objects a theory refers to are treated *as if* they exist independently of what a mathematician can explicitly define or construct.

By the 1920s, Russell’s (simplified) type theory and Zermelo’s set theory had largely converged on this same methodology.

Summary - Response to the Paradoxes

The paradoxes did **not** kill modern mathematics.

- They forced a *clarification*: existence via **consistency**, not naive comprehension
- Poincaré’s predicativism and Russell’s type theory tried to *restrain* the new methods
- Zermelo’s set theory — soon ZFC — *codified* the new methods instead.

A New Kind of Critique

1921: Weyl publishes a paper in the *Mathematische Zeitschrift* — a disciple of Hilbert’s — openly siding with **Brouwer**, diagnosing a “crisis in the foundations.”

- Weyl’s paper: deliberately a *propaganda pamphlet*, “to rouse the sleepers”
- **Hilbert’s reply**, same year: Brouwer and Weyl are attempting a “**putsch**” — a “dictatorship à la Kronecker”

The foundational debate turns into open battle: Hilbert’s classical mathematics vs. Brouwer’s constructive “revolution.”

What is Brouwer’s “Revolution”?

Up to 1920, the debate concerned: real numbers, impredicativity, and the strong existential assumptions of set theory (esp. AC).

Brouwer went further:

- He put **classical logic itself** on trial: quantifiers over infinite domains, and the **principle of excluded middle** (PEM)
- Poincaré, Klein, and others had said logic is not the whole of mathematics; Brouwer said mathematics is *prior to and independent of* logic and language altogether.

Intuitionism — First Principles

- For Brouwer, mathematical objects are **mental constructions**; mathematics does not depend on language or logic to exist
- Classical logic (including PEM) was abstracted from experience with **finite** sets — and wrongly extrapolated to infinite ones
- PEM: $p \vee \neg p$ — “either p or not p ” — always holds, classically

Brouwer’s claim: this is *not* a law of thought, but an unjustified generalization.

The Constructive Standard

Consequence: **proof by contradiction is invalid**.

- Classically: assume $\neg p$, derive a contradiction, conclude p
- Constructively: this only shows $\neg\neg p$ — not the same as a *construction* of p

A Different Notion of Truth

For a constructivist:

- “ p is **true**” means: we possess a proof of p by stringent constructive methods
- “ p is **false**” means: we possess a counterexample to p

To a constructivist, the statement “ **is normal**” currently has *no truth value* — not true, not false, not “unknown-but-determinate.” Truth is not timeless; it is made.

Rebuilding Analysis: Choice Sequences

Brouwer (1918) replaces Cantorian sets with constructive substitutes: **spreads** and **species**.

- A **choice sequence**: a sequence of rationals not required to be fully specified in advance
 - may be generated by a fixed rule (e.g. Newton’s method for $\sqrt{2}$)
 - or by choices the mathematician makes freely, *over time*, subject to constraints

Intuitionistic mathematics is “*mathematics in the making*” — unlike the timeless, fully-determinate objects of classical analysis.

Spreads and a New Continuum

- A **spread** is a law regulating which choice sequences are admissible — the basis for Brouwer’s reconstruction of $\mathbb{R}$
- The continuum becomes a genuinely **continuous domain**, not a set of pre-existing points
 - an echo of Aristotle: the continuum is *not* made up of unextended points

A function on a spread must be determined from a **finite initial segment** of any choice sequence — this has a startling consequence.

The Startling Consequence

Every intuitionistically total function on $\mathbb{R}$ is (uniformly) **continuous**.

- The classical step function

$$f(x) = \begin{cases} 0 & x \leq 0 \\ 1 & x > 0 \end{cases}$$

is **not a legitimate function** for Brouwer

- Reason: deciding $x > 0$ vs. $x \leq 0$ is not always possible — not *decidable* from a finite stage of the sequence

(Anticipates the later, independent discovery that all computable real functions are continuous — Turing and others.)

The Cost of Intuitionism

Rebuilding analysis this way is hard. Weyl, who once championed Brouwer’s program, wrote in 1925:

“The mathematician watches with pain the largest part of his towering theories dissolve into mist before his eyes.”

Brouwer himself, unfazed, replied (1933):

“The spheres of truth are less transparent than those of illusion.”

Weyl abandoned intuitionism shortly after. But Hilbert had another answer in mind.

Hilbert’s Program

Hilbert (1928): to “**eliminate from the world once and for all the skeptical doubts**” about classical mathematics.

- Strategy: use **Kroneckerian** (finitary, constructive) means to justify **anti-Kroneckerian** (classical) methods
- Key move: turn proofs into **formal, syntactic objects** — finite strings of symbols manipulated by explicit rules
- A proof becomes something a **finitary, combinatorial** argument can reason *about*, from the outside — **metamathematics**

The Four Steps of the Program

1. Find suitable **axioms and primitive concepts** for a theory T (e.g. the real numbers)
2. Find axioms and rules for **classical logic** — syntactic rules turning propositions into new propositions
3. **Formalize T** : propositions become strings of symbols, proofs become sequences of strings obeying formal rules
4. Give a **finitary** proof that no string expressing a contradiction can ever be the last line of a proof

If (iv) succeeds for T : consistency is established by means even a strict finitist should accept.

Finitism: What Counts as “Given”

Hilbert’s answer to Poincaré’s 1905 circularity worry (a consistency proof for arithmetic would need induction):

- What is genuinely, intuitively given: **signs and formulas** themselves, as concrete finite objects
- Metamathematical reasoning about them needs only a *very weak* form of induction — much weaker than full arithmetical induction
- Finitary mathematics needs **no further justification** — it is where justification bottoms out

Progress and Limits (1920s)

- Propositional logic: proved **consistent and complete**
- First-order (predicate) logic: proved **complete** — Gödel’s dissertation, 1929
- Ackermann, von Neumann: consistency results for **subsystems** of arithmetic (1928–1930)
- By 1930 Hilbert believed full consistency of arithmetic was within reach

A crucial hidden assumption: that the axiomatic systems in question would also be **complete**.

“Formalism” — a Slightly Misleading Name

- The label comes from Hilbert’s *method*: formalize a theory, study its formal proof-structure
- But Hilbert did **not** think mathematics is a meaningless game with symbols
 - he consistently stressed the *meaningfulness* of informal mathematical statements
- Unlike intuitionism, formalism was never a full philosophy of mathematics — it was a **method** for justifying the classical one

A Personal Crisis, Too

The dispute was not only intellectual.

- Hilbert and Brouwer: strong-willed, brilliant, and deeply incompatible personalities
- **1928**: Hilbert moves to remove Brouwer from the editorial board of *Mathematische Annalen*
 - Einstein: the move is “unwise” and wants no part of it
 - The board is dissolved and reconstituted without Brouwer — the journal loses Einstein and Carathéodory

- Brouwer, disturbed, withdraws from the scene for years — as political tensions fade, so does the sense of “crisis”

Looking Ahead

Hilbert’s program looked to be within reach by 1930 — one more push on the consistency of arithmetic.

Things turned out otherwise...